Linear groups and regular polytopes: the complete thesis classification

Human Compression project: the N012 principal recovery

This paper assembles the principal results of I. A. Markovskaya's 2025 thesis [T], in its fixed eighty-page edition. We acquire the complete companion proofs [I,U,L,Z,H,Q], rather than repeat their matrix calculations. One structural clause in Corollary 4.2.1 of [T] requires correction; its polytope conclusion remains valid. The classification below includes all twelve numbered principal theorems and corollaries and both six-row tables of [T].

Every dimension is at least two. An admissible group is generated by three involutions with a commuting pair, with the source convention allowing repetitions. A rank-three regular-polytope group instead requires an actual string C-marking, including the intersection property and distinct nonidentity generators. For odd q write

$$H_n(q) = \{g \in \mathrm{GL}_n(q) : \det g \in \{1, -1\}\}, \qquad Q_n(q) = H_n(q)/Z(\mathrm{SL}_n(q)).$$

Thus $Q_n(q)$ is the source's $\mathrm{PGL}_n^{\pm 1}(q)$. It need not equal the quotient by the full centre of $H_n(q)$.

1 The principal classifications

Theorem 1. *For $n \geq 2$, the integral groups satisfy*

$$\mathrm{GL}_n(\mathbb{Z}) \text{ admissible} \iff n \geq 5, \qquad \mathrm{PGL}_n(\mathbb{Z}) \text{ admissible} \iff n = 2 \text{ or } n \geq 5.$$

Over a finite field the following table gives exactly the admissible parameters. The regular-polytope parameters are the same except that $\mathrm{GL}_2(2)$ and $\mathrm{PGL}_2(2)$ are omitted.

group	*admissible parameters*
$\mathrm{GL}_n(q)$	$q = 2,\ n = 2$ *or* $n \geq 5$*; or* $q = 3,\ n \geq 5$
$\mathrm{PGL}_n(q)$	$n = 2$*; or* $n \geq 4,\ \gcd(n, q-1) = 2$*;* *or* $n \geq 5,\ \gcd(n, q-1) = 1$
$H_n(q)$*, q odd*	$n \geq 5$
$Q_n(q)$*, q odd*	$n = 2,\ q \neq 9$*; or* $n \geq 4$

For odd $q = p^m$ and $n \geq 5$, $H_n(q)$ has a rank-three regular polytope of type $[4p, n]$ or $[4p, 2n]$.

Proof. The full integral argument is [I]. It gives unit-root words for every $n \geq 5$, treats both degree-two groups directly, and excludes degrees three and four by the explicitly retained finite quotients. It is precisely [T, Theorem 2.2.1].

The two finite general-linear admissibility classifications are the theorems labelled `gl` and `pgl` in [L]. Their parameter sets agree exactly with [T, Theorems 3.2.1 and 3.3.1], and the two tables in [L] recover every row of [T, Tables 1 and 2]. The integral premise formerly retained by [L] is now supplied by [I]. This is acyclic: [I] uses the separate exclusions $\mathrm{PSL}_3(2), \mathrm{PSL}_4(2)$, not the finite general-linear classification that subsequently uses integral reduction.

For $q = 3$, $\mathrm{GL}_n(3) = H_n(3)$, so the actual marking and intersection proof of [H] prove [T, Theorem 3.2.2] in every positive dimension. For $q = 2$, the positive groups are simple and the explicitly retained admissibility and intersection criteria apply. The remaining admissible group

$S_3 = \mathrm{GL}_2(2)$ has no distinct commuting involutions, so it has no rank-three C-marking. The projective polytope statement is proved in [L, Section 5], with $\mathrm{PGL}_2(3) = S_4$ handled separately from the almost-simple cases; the same S_3 exclusion completes [T, Corollary 3.3.1]. These deductions are written out in [Q].

The first theorem of [H] gives the complete H classification and types, namely [T, Theorems 4.2.1 and 4.2.2]. It acquires the full Scott obstruction of [Z] in degrees two, three and four. The second theorem of [H] supplies actual C-markings for all scalar quotients in dimensions at least five. In dimension four [Z] proves the Q assertion over every odd field, with both centre cases treated. Finally [Q] proves the precise small quotient cases: $Q_2(q)$ is positive exactly when $q \neq 9$, and every $Q_3(q)$ is negative. Its central direct products use the actual perfect-group lift of [Z]. This proves [T, Theorem 4.3.1 and Corollary 4.3.1], including their negative assertions. □

2 The three quotient corollaries

Put $Z_1 = Z(H_n(q))$, $Z_2 = Z(\mathrm{SL}_n(q))$, $d = \gcd(n, q-1)$ and $e = (q-1)/d$.

Theorem 2. *For odd $q = p^m$ and $n \geq 5$, the following conclusions hold.*

1. *If n is even, the full-centre quotient $H_n(q)/Z_1$ is $\mathrm{PSL}_n(q)$ when e is even. When e is odd, it is the index-two extension $\mathrm{PSL}_n(q) \rtimes C_2$ embedded in $\mathrm{PGL}_n(q)$, with complement represented by $\mathrm{diag}(-1, 1, \ldots, 1)$. In both cases it has a rank-three regular polytope of type $[4p, n]$.*
2. *If n is odd, $H_n(q)/Z_1 \cong \mathrm{PSL}_n(q)$ has a rank-three regular polytope of type $[4p, n]$.*
3. *If n is odd, $H_n(q)/Z_2 \cong \mathrm{PSL}_n(q) \times C_2$ has a rank-three regular polytope of type $[4p, n]$ or $[4p, 2n]$.*

The first clause corrects the structural part of [T, Corollary 4.2.1]; the second and third recover Corollaries 4.2.2 and 4.2.3 as stated.

Proof. The centre theorem of [Z] proves exactly these group isomorphisms. When e is even, $|Z_1 : Z_2| = 2$, and otherwise $Z_1 = Z_2$. Odd n always has even e, since $-I$ has determinant -1. The second theorem of [H], applied first to Z_1 and then to Z_2, proves the asserted types by an actual quotient intersection calculation. In its notation $|AC| = 4p$ and $|\langle A, C\rangle| = 8p$; this dihedral subgroup meets the scalar kernel trivially. The monomial product BC has order n or $2n$, and has order exactly n after dividing by the full centre. No cyclic-normal-subgroup criterion is applied to the direct product in the third clause. □

For clarity, the printed even-dimension assertion $Z_1 = Z_2$ on [T, p. 69] is false: at $n = 6, q = 5$, $2I$ has determinant -1, $|Z_1| = 4$ and $|Z_2| = 2$. Thus the structural clause of its numbered corollary genuinely changes. Its type and existence assertions do not change. The corrected classification is the principal disposition recorded here.

3 Attribution and the exact boundary

All source principal results are attributed to Markovskaya and the collaborators named in her thesis. Its pages 6–8 credit the integral theorem jointly with the supervisor, the difficult five- and six-dimensional triples to the author, the finite general-linear negative cases jointly with the supervisor, and the explicit positive constructions and Chapter 4 results to the author. The companion papers distinguish the source matrices and Scott method from the complete common root arguments, repaired identities, centre calculation, central lift and uniform intersection proofs supplied in this project. No general priority claim is made.

The precise primary ledger has twelve rows. Supporting lemmas and survey references are recorded separately. In particular the rank-two Dickson–Levchuk assertions are not premises of the

new root argument or the small quotient proof. The external inputs actually retained are the stated simple-linear admissibility classification, Sjerve–Cherkassoff's PGL_2 theorem, Conder–Oliveros and the usual group–polytope correspondence. Their full external proof and software costs remain unknown.

The corrections in [Z,H] are retained explicitly: the printed degree-four matrix on [T, p. 72] is not a projective involution, and the seven-dimensional word on p. 67 needs the recorded square. The former defect was already found in the project's phase 22 audit. The replacement uses corrected witnesses. For the finite $\mathrm{GL}_6(3)$ polytope it does not claim that its marking is the original integral marking with the source's pair orders $24, 12$. It proves the same principal existence conclusion.

The assembly uses no new finite search, no new external classification and no new Lean declaration. Complete companion manuscripts and their exact validation and distinct review records are required dependencies. This paper proposes one corrected principal recovery of N012; independent whole-source review and publication acceptance remain separate. It asserts no new open-conjecture resolution and no compression factor.

References

[1] [T] I. A. Markovskaya, *Linear groups over finite fields and over the ring of integers as automorphism groups of regular polytopes of rank 3*, thesis, Siberian Federal University, 2025, fixed eighty-page edition; supervisor Ya. N. Nuzhin.

[2] [I] *Integral linear groups generated by three involutions.*

[3] [U] *Root parameters and fixed-space obstructions.*

[4] [L] *Three involutions in finite general linear groups.*

[5] [Z] *Central quotients of the determinant-sign subgroup.*

[6] [H] *Involutions and intersections in groups with determinant* ± 1, complete accepted second edition; the portable copy clarifies the product and dihedral subgroup orders explicitly.

[7] [Q] *The remaining quotients and the linear polytope classifications.*